lec 18 ① linear function approximation (simulator setting) LSVI

lec 19 ② ... (exploration setting) LSVI-UCB

Simulator Setting

we know feature map $\phi(\cdot,\cdot)$ ϕ is assumed given

can query any (s, a, h) and obtain r. and sample s'

how to query (s, a, h) ? maybe infinite

$r_h(s, a) = \theta_h^T \phi(s, a)$ $\mathbb{P}_h(\cdot \mid s, a) = \langle \phi(s, a), \mu_h^T(\cdot) \rangle$

Ideally. if we have $\underbrace{\{(s_i, a_i)\}_{i=1}^{d}}_{\text{core set}}$ s.t. $\phi(s_i, a_i) \approx e_i$

the query (s_i, a_i) $\Leftrightarrow$ query ith coordinate of θ_h/transition

Core Set K.

$\{(s_i, a_i)\}_{i=1}^{d}$ denote $\psi_i(s_i, a_i)$ $\Psi_i = \{\psi_1, \psi_2, \dots \psi_d\}$

properties.

① Ψ is full rank

② $\forall (s, a)$, $\|\Psi^{-1}\phi(s, a)\|_1 \le L$

Algorithm : LSVI with core set

Input n

for all $h \in [H]$, all $(s, a) \in K$ query simlators & collect data

query (s_0, h) and collect n samples $\{s_1', s_2', \dots s_n'\}$

put tuples $\{(s, a, s_i')\}_{i=1}^{n}$ into dataset D_n

for $h = H, H-1, \dots 1$ LSVI

$$\hat{w}_h = \underset{w}{\operatorname{argmin}} \sum_{(s,a,s') \in D_n} \left(w^T \phi(s,a) - \underbrace{r(s,a)}_{} - \hat{V}_{h+1}(s')\right)^2$$

for simplicity

assume reward is known

$$\hat{Q}(s,a) = \hat{w}_h^T \phi(s,a)$$

$$\hat{V}_h(s) = \max_a \hat{Q}_h(s,a)$$

return policy $\hat{\pi}$, s.t. $\hat{\pi}_h(s) = \operatorname{argmin}_a \hat{Q}_h(s,a)$

remark 1. feature of core set $\{\psi_1, \psi_2, \dots \psi_d\}$ span over d-dimention

remark 2. any feature vector $\phi(s,a)$ can represent by linear combination of core set with small weights:

$$\phi(s,a) = \sum_{i=1}^{d} w_i \psi_i = \Psi \vec{w} \Rightarrow \vec{w} = \Psi^{-1} \phi(s,a)$$

$$\|\vec{w}\|_1 = \|\Psi^{-1} \phi(s,a)\|_1 \le L$$

Example

simplex feature space $\{\phi(s,a) \mid (s,a) \in S \times A\} = \Delta_d$

then $\{(s_i, a_i)\}_{i=1}^{d}$ s.t. $\phi(s_i, a_i) = e_i$ give a core set

$\Psi = (e_1 e_2 \dots e_d) = \underset{\text{identity matrix}}{I}$ $\qquad \|\Psi^{-1} \phi(s,a)\|_1 = \|\phi(s,a)\|_1 = 1$

Th1.

if we choose $n \geq C\frac{L^2H^2\iota}{\varepsilon^2}$, where $\iota = \log\frac{4d}{p}$,

then w.p. $\geq 1-p$, the output $\hat{\pi}$ of LSVI with core set would be ε-optimal.

Cor1.

the sample complexity for a ε-optimal policy with high probality is $nHd = c\frac{H^5 d L^2 \iota}{\varepsilon^2}$

the size of core set

lem2.

$$V_1^*(s_1) - \hat{V}_1(s_1) \leq \mathbb{E}_{\pi^*}\left[\sum_{h=1}^{H}[(\mathbb{P}_h - \hat{\mathbb{P}}_h)\hat{V}_{h+1}](s_h, a_h) \,\middle|\, s_1\right]$$

pf: similar to lec5

$$Q_h^*(s,a) - \hat{Q}_h(s,a) = (\mathbb{P}_h V_{h+1}^*)(s,a) - (\hat{\mathbb{P}}_h \hat{V}_{h+1})(s,a)$$

$$= [(\mathbb{P}_h - \hat{\mathbb{P}}_h)\hat{V}_{h+1}](s,a) + [\mathbb{P}(V_{h+1}^* - \hat{V}_{h+1})](s,a)$$

$$V_h^*(s) - \hat{V}_h(s) = \max_a Q_h^*(s,a) - \max_a \hat{Q}_h(s,a)$$

$\pi^*(s)$

$$\leq Q_h^*(s, \pi^*(s)) - \hat{Q}_h(s, \pi^*(s))$$

$$\Rightarrow \underbrace{V_h^*(s) - \hat{V}_h(s)}_{h \text{ step}} \leq \mathbb{E}_{\pi^*}[[(\mathbb{P}_h - \hat{\mathbb{P}}_h)\hat{V}_{h+1}](s_h, a_h) \mid s_h = s]$$

$$+ \mathbb{E}_{\pi^*}[\underbrace{(V_{h+1}^* - \hat{V}_{h+1})}_{h+1 \text{ step}}(s_{h+1}) \mid s_h = s]$$

use recursion:

$$V_1^*(s) - \hat{V}_1(s) \leq \mathbb{E}_{\pi^*}\left[\sum_{h=1}^{H}[(\mathbb{P}_h - \hat{\mathbb{P}}_h)\hat{V}_{h+1}](s_h, a_h) \,\middle|\, s_1\right] \quad \blacksquare$$

pf of Th1.

$$\hat{w}_h = \underset{w}{\operatorname{argmin}} \sum_{(s,a,s')\in D_h} \left(w^T\phi(s,a) - r(s,a) - \hat{V}_{h+1}(s')\right)^2$$

$$= \Lambda^{-1} \sum_{i=1}^{N} \phi_i \left(r_h(s_i, a_i) + \hat{V}_{h+1}(s_i') \right)$$

where $\Lambda = \sum_i \phi_i \phi_i^T \qquad \phi_i = \phi(s_i, a_i) \qquad N = nd$

$$\hat{Q}_h(s,a) = \phi(s,a)^T \hat{w}_h$$

linear . $\phi_i^T \theta_h$

$$= \phi(s,a)^T \Lambda^{-1} \sum_{i=1}^{N} \phi_i \left(r_h(s_i, a_i) + \hat{V}_{h+1}(s_i') \right)$$

$$= \phi(s,a)^T \underset{I}{\Lambda^{-1}} \underbrace{\sum_{i=1}^{N} \phi_i \phi_i^T}_{\Lambda} \theta_h + \phi(s,a)^T \Lambda^{-1} \sum_{i=1}^{N} \phi_i \hat{V}_{h+1}(s_i')$$

$$= \phi(s,a)^T \theta_h + \int \underbrace{\phi(s,a)^T \Lambda^{-1} \sum_{i=1}^{N} \phi_i}_{\text{scalar}} \delta(s', s_i') \hat{V}_{h+1}(s') \, ds'$$

$$= r(s,a) + (\hat{\mathbb{P}} \hat{V}_{h+1})(s,a)$$

where $\hat{\mathbb{P}}_h(\cdot \mid s,a) := \sum_i \phi(s,a)^T \Lambda^{-1} \phi_i \, \delta(\cdot, s_i')$

Inspect error

$$\left[(\mathbb{P}_h - \hat{\mathbb{P}}_h)\hat{V}_{h+1}\right](s,a) = \phi(s,a)^T \int \hat{V}_{h+1}(s') \, d\vec{\mu}(s')$$

$$- \phi(s,a)^T \Lambda^{-1} \sum_{i=1}^{n} \phi_i \hat{V}_{h+1}(s_i')$$

$$\left(\Lambda^{-1} \sum_{i=1}^{n} \phi_i \phi_i^T = \mathbb{I} \right)$$

$$= \phi(s,a)^T \Lambda^{-1} \sum_{i=1}^{n} \phi_i \left[\phi_i^T \int \hat{V}_{h+1}(s') \, d\vec{\mu}(s') - \hat{V}_{h+1}(s_i') \right]$$

$$= \phi(s,a)^T \Lambda^{-1} \sum_{i=1}^{n} \phi_i \left[\mathbb{E}(\hat{V}_{h+1}(s') \mid s_i, a_i) - \hat{V}_{h+1}(s_i') \right]$$

recall $\phi_i = \phi(s_i, a_i)$, s_i' are data collected in h step

$N = nd$, with n samples per feature in core set

$$\Lambda = \sum_{i=1}^{n} \phi_i \phi_i^T = n \sum_{j=1}^{d} \psi_j \psi_j^T = n \Psi \Psi^T$$

Hoeffding's $\qquad |\varepsilon_j| \le c H \sqrt{\frac{L}{n}}$

union bound

$$[(\mathbb{P}_h - \hat{\mathbb{P}}_h)\hat{V}_{h+1}](s,a) = \phi(s,a)^T (\Phi\Phi^T)^{-1} \sum_{j=1}^{d} \phi_j \underbrace{\left[\mathbb{E}(\hat{V}_{h+1}(s')|\psi_j - \frac{1}{n}\sum_{\ell=1}^{n} \hat{V}_{h+1}(s'_\ell)\right]}$$

$$= \phi(s,a)^T (\Phi\Phi^T)^{-1} \sum_{j=1}^{d} \phi_j \varepsilon_j$$

$$= (\Phi^{-1}\phi(s,a))^T \vec{\varepsilon}$$

Holder's inequality

if $\frac{1}{p} + \frac{1}{q} = 1$, then

$\|fg\|_1 = \|f\|_p \|g\|_q$

$$\leq \underbrace{\|\Phi^{-1}\phi(s,a)\|_1}_{\leq L} \|\vec{\varepsilon}\|_\infty$$

$$\leq CLH\sqrt{\frac{L}{n}}$$

$$V_1^*(s_1) - \hat{V}_1(s_1) \leq \mathbb{E}_{\pi^*}\left[\sum_{h=1}^{H-1} [(\mathbb{P}_h - \hat{\mathbb{P}}_h)\hat{V}_{h+1}](s_h, a_h) | s_1\right]$$

$$\leq CLH^2\sqrt{\frac{L}{n}} \leq \varepsilon$$

$$\therefore n \geq C' \frac{L^2 H^4 \iota}{\varepsilon^2}$$

lec19 ① LSVI-UCB

② general function approximation, FQI

Exploration Setting

no longer able to query "core set", we have to

① generalize knowledge from non-core-set data

② scheme to try diverse data (done by UCB)

Idea: incorporate UCB into LSVI

LSVI-UCB:

for episode $k = 1, 2, \ldots K$ do

receive initial state s_1^k ← fixed

for step $h = H, \ldots, 1$ do $\quad \phi_h^\tau = \phi(s_h^\tau, a_h^\tau)$

LSVI

$$\Lambda_h \leftarrow \sum_{\tau=1}^{k-1} \phi_h^\tau (\phi_h^\tau)^T + \lambda I$$

with bonus

$$w_h \leftarrow \Lambda_h^{-1} \sum_{\tau=1}^{k-1} \phi_h^\tau \left[r_h(s_h^\tau, a_h^\tau) + \max_a Q_{h+1}(s_{h+1}^\tau, a) \right]$$

$$Q_h(\cdot,\cdot) \leftarrow \min\{ \underset{\text{truncation}}{H}, \underbrace{w_h^T \phi(\cdot,\cdot)}_{\text{linear}} + \underbrace{\beta \sqrt{\phi(\cdot,\cdot)^T \Lambda_h^{-1} \phi(\cdot,\cdot)}}_{\text{bonus}} \}$$

execute greedy policy

for step $h = 1, 2, \dots H$ do

take action $a_h^k \leftarrow \arg\max_a Q_h(s_h^k, a)$, observe s_{h+1}^k

Comparison with Tabular Setting

bonus: $\sqrt{\frac{1}{N^k(s_i, a_i)}}$ $\quad \phi \in \{e_i \mid i \in [d]\}$ $\quad d = SA$

$$\sqrt{e_i^T \Lambda^{-1} e_i} = \sqrt{\frac{1}{N_i}} \quad \Lambda = \sum_i \underset{\#\text{ of times observing } e_i}{N_i} e_i e_i^T$$

In Linear Setting

$$\sqrt{\phi^T \Lambda^{-1} \phi} \approx \sqrt{\frac{1}{\tilde{n}(\phi)}} \leftarrow \text{"effective" \# of samples in } \phi$$

$\Rightarrow$ we can prove LSVI-UCB $\sqrt{d^3 H^3 T}$ regret

General Function Approximation.

Q_h-functions belong to general function classes $\{\mathcal{F}_h\}$

infinite # of states, finite # of actions

What structure allow sample-efficient RL?

the following algorithm may not be computationally efficient X

Denote Bellman Operator $\mathcal{T}_h : \mathbb{R}^{S \times A} \longrightarrow \mathbb{R}^{S \times A}$

$$(\mathcal{T}_h f)(s,a) = r_h(s,a) + \mathbb{E}_{s'\sim \mathbb{P}(\cdot|s,a)} \max_{a'} f(s',a')$$

Assumption 1. (realizability) $Q_h^* \in \mathcal{F}_h$, $\forall h$ $\quad (Q_{H+1}^* = 0 \in \mathcal{F}_{H+1})$

Assumption 2. (closeness) $\forall f \in \mathcal{F}_h$, $\mathcal{T}_{h-1} f \in \mathcal{F}_{h-1}$ $\quad (0 \in \mathcal{F}_{H+1})$

In linear MDP, both are true

Assumption 3. (access to represention data)

provided data $\{(s_h^{(i)}, a_h^{(i)})\}_{i=1}^N$ for all $h \in [H]$

where $(s_h^{(i)}, a_h^{(i)}) \overset{i.i.d.}{\sim} \mu_h$ s.t. $\forall$ policy π, $\forall (s,a)$, $\dfrac{\mathbb{P}_h^\pi(s,a)}{\mu_h(s,a)} \le C$

probality of (s,a) at step h following π

In tabular MDP with simulator setting

$\mu_h(s,a) = \frac{1}{SA}$

$\mathbb{P}_h^\pi(s,a) \le 1$

$\rightarrow$ setting $C = SA$, satisfy assumption 3

denote the loss:

$$\mathcal{L}_{D,h}(f_h, f_{h+1}) = \frac{1}{N} \sum_{(s,a,s')\in D} \Big(\underbrace{f_h(s,a)}_{Q_h} - r_h(s,a) - \underbrace{\max_{a'} f_{h+1}(s',a')}_{V_{h+1}} \Big)^2$$

$$\mathbb{E}[\mathcal{L}_{D,h}(f_h, f_{h+1})] = \mathbb{E}_{\substack{(s,a)\sim\mu_h \\ s'\sim\mathbb{P}(\cdot|s,a)}} [f_h(s,a) - r_h(s,a) - V_{h+1}(s')]^2$$

only depend on (s,a) $\qquad \mathbb{E}(\cdot|s,a) = 0$

$$= \mathbb{E}_{(s,a,s')} [\underbrace{f_h(s,a) - r_h(s,a) - (\mathbb{P}_h V_{f_{h+1}})(s,a)} + \underbrace{(\mathbb{P}_h V_{f_{h+1}})(s,a) - V_{f_{h+1}}(s')}]^2$$

$$= \mathbb{E}_{(s,a)\sim\mu} [f_h(s,a) - r_h(s,a) - \mathbb{P}_h V_{f_{h+1}}(s,a)]^2 + \mathbb{E}_{(s,a,s')} [\mathbb{P}_h V_{f_{h+1}}(s,a) - V_{f_{h+1}}(s')]^2$$

$$= \mathbb{E}_{(s,a)\sim\mu} [(f_h - \mathcal{T}_h f_{h+1})(s,a)]^2 + \mathbb{E}_{(s,a)\sim\mu} \mathrm{Var}[V_{f_{h+1}}(s') | s,a]$$

$$= \underbrace{\| f_h - \mathcal{T}_h f_{h+1} \|_{2,\mu}^2}_{\text{Bellman error under distribution } \mu} + \underbrace{\mathbb{E}_{(s,a)\sim\mu} \mathrm{Var}[V_{f_{h+1}}(s') \mid s,a]}_{\text{variance independent of } f_h}$$

FQI (Fitted Q-Iteration)

Initial $\hat{f}_{H+1} = 0$

for $h = H, \dots, 1$

$$\hat{f}_h \leftarrow \underset{f \in \mathcal{F}_h}{\arg\min} \mathcal{L}_{D,h}(f, \hat{f}_{h+1})$$

note. 1. LSVI is a special case of the update ($\mathcal{F}_h$ linear)

2. in general, the minimizer $\hat{f}_h$ may not be computed efficiently

In the limit : $n \to \infty$

$$\hat{f}_h = \underset{f \in \mathcal{F}}{\arg\min} \| f - \mathcal{T}_h \hat{f}_{h+1} \|_{2,\mu}^2$$

since $\hat{f}_{h+1} \in \mathcal{F}_{h+1}$, by closeness property, $\mathcal{T}_h \hat{f}_{h+1} \in \mathcal{F}$

$\to$ therefore $\hat{f}_h = \mathcal{T}_h \hat{f}_{h+1}$ is global minimizer

conversely $\| f - \mathcal{T}_h \hat{f}_{h+1} \|_{2,\mu}^2 = 0 \quad \to \quad f = \mathcal{T}_h \hat{f}_{h+1}$

hence $\hat{f}_h = \mathcal{T}_h \hat{f}_{h+1}$, $\forall h$

$\hat{f}_{H+1} = 0 = Q^*_{H+1}$, by bellman eq. $\hat{f}_h = Q^*_h \ \forall h$